

Optimisation method landscape: convergence rate vs.\ information cost

Convergence rate (quality)

Quadratic
Superlinear
Linear $\propto k$
 $(1/k^2)$
 $(1/k)$
 $(1/\sqrt{k})$

Derivative-free

First-order

Second-order

Newton (unit step)

Newton (damped)

L-BFGS

Nesterov

Adam

Polyak step

Heavy Ball

Adagrad

RMSprop

Projected GD

Conjugate GD

SGD

FD-GD (good delta)

FD-GD (poor delta)

Nelder-Mead

Nesterov random search

Grid search

FE only

FD grad

Exact grad

Diagonal Hess.

Full Hessian

Information required per iteration (cost)

